EXPERIMENT 3: STUDY OF NETWORK CABLES

AIM:

To study network cable types and understand how to prepare straight-through and crossover Ethernet cables.

PROCEDURE:

1. Collect UTP cable, RJ-45 connectors, a crimping tool, wire cutter and cable tester.

2. Strip approximately 1.5 cm of the outer cable jacket without damaging the inner wires.

3. Untwist and arrange the eight wires in the required T568A or T568B order.

4. Trim the wires evenly and insert them into an RJ-45 connector, ensuring each wire reaches the end.

5. Crimp the connector and repeat at the other end.

6. For a straight-through cable, use the same standard at both ends; for a crossover cable, use T568A at one end and T568B at the other.

7. Test the cable with a cable tester.

8. Avoid pulling or sharply bending the cable while preparing the connectors.
9. Ensure that the jacket is held securely by the connector and that no copper conductor is exposed outside it.
10. Record the cable tester's pin sequence and check for open, shorted or incorrectly ordered wires.

CODE:

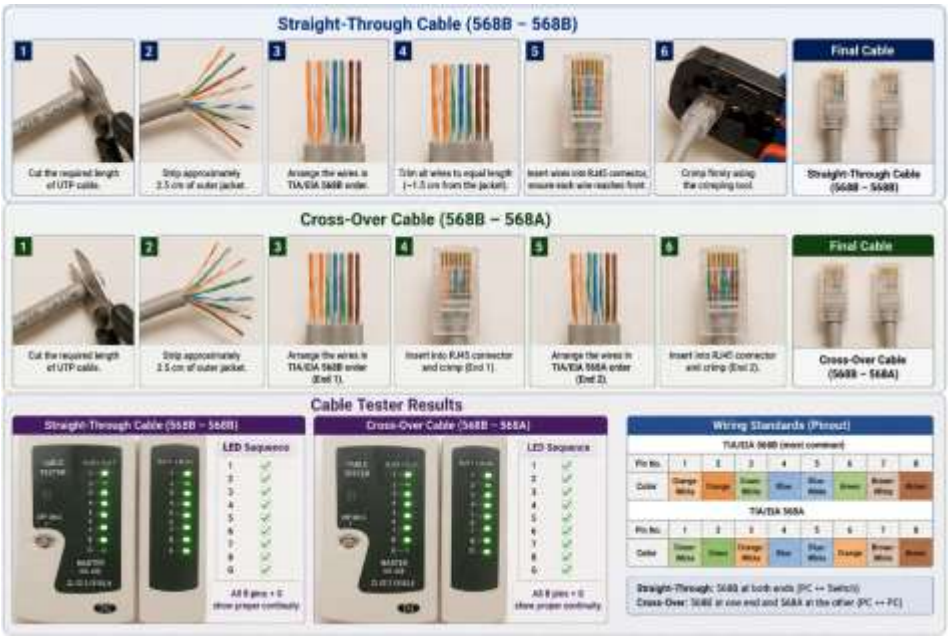
```
T568A pin order:
1. White-Green
2. Green
3. White-Orange
4. Blue
```

5. White-Blue
6. Orange
7. White-Brown
8. Brown

T568B pin order:
1. White-Orange
2. Orange
3. White-Green
4. Blue
5. White-Blue
6. Green
7. White-Brown
8. Brown

Straight-through: T568B at both ends
Crossover: T568A at one end and T568B at the other

OUTPUT:



RESULT:

Thus, straight-through and crossover cable wiring standards were studied, and the cable-testing procedure was understood.